8^{th} Grade Unit 3 Assessment Guide

General Information	Unit 3 introduces students to scientific notation by building on the Laws of Exponents students learned in grade 7 while also introducing students to negative powers of 10. Unit 1 supports students in their recall of the laws of exponents. In 5^{th} grade, students learned to write multi-digit numbers with decimals to the thousandths using expanded form (MA.5.NSO.1.2). Students who struggle with connecting place value to scientific notation may benefit from revisiting writing numbers using expanded form. This could be more effective to help students who struggle with negative powers of 10. For example, writing 0.004 in expanded form is $4 \times \frac{1}{1000}$. Extending the expanded form to scientific notation may help students who struggle. By the end of grade 7, students were expected to have been fluent with operations with rational numbers. Units 1 and 2 helps to reorient students to rational numbers but it is in this unit where teachers may discover which students struggle with performing operations with rational numbers. Grade 7, Unit 3 may be a resource teachers can use to support students who struggle with operations with rational numbers. Students will build on the Laws of Exponents in Unit 4 as they move beyond powers of 10. Students have been evaluating exponents since Grade 6 and applying Laws of Exponents with whole number exponents since Grade 7. Grade 8 is an accumulation of this knowledge. Teachers should use this assessment to determine if students need further support on these laws before students apply multiple laws in Unit 4. Extremely large or extremely small numbers can be difficult for some students. This could be because these are not numbers students use on a daily basis.
Calculator Information	The questions on the unit assessment are no calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.8.NSO.1.3
☐ $-\left(\frac{1}{10}\right)^2$ ☐ $\frac{1}{-20}$ ☐ $\frac{1}{10^2}$ ☐ -20 ☐ $(-2)^{-10}$ ☐ $\frac{1}{10} \cdot \frac{1}{10}$ ☐ $-\left(\frac{1}{10} \cdot \frac{1}{10}\right)$		Recommended Scoring (8 Total Points)	Consider giving students 4 points for each correct answer and deduct 2 points for each incorrect answer. A student who selects all of the answers should receive no points, and a student should not receive negative points
Question	2a	Benchmark(s)	MA.8.NSO.1.4
Scientific Notation	5.03×10^5	Recommended Scoring (4 Total Points)	No partial credit should be given.
Question	2b	Benchmark(s)	MA.8.NSO.1.4
Scientific Notation	3.2×10^6	Recommended Scoring (4 Total Points)	No partial credit should be given.
Question	3	Benchmark(s)	MA.8.NSO.1.4
Comparison	20 times larger	Recommended Scoring (4 Total Points)	No partial credit should be given.

Question	4	Benchmark(s)	MA.8.NSO.1.4
Standard Form — 1,390,000,000,000,000,000,000 feet		**Recommended Scoring (4 Total Points)**	No partial credit should be given.
Question	**5**	**Benchmark(s)**	MA.8.NSO.1.4
Standard Form — 0.000008056 miles per second		**Recommended Scoring (4 Total Points)**	No partial credit should be given.
Question	**6a**	**Benchmark(s)**	MA.8.NSO.1.4
Comparison — 4 times smaller		**Recommended Scoring (4 Total Points)**	No partial credit should be given.
Question	**6b**	**Benchmark(s)**	MA.8.NSO.1.6
Total — 3.4858×10^6 mg		**Recommended Scoring (4 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	6c	Benchmark(s)	MA.8.NSO.1.6
Difference: 1.34142×10^{7} mg		Recommended Scoring (4 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
		Additional Notes	Students may use significant digits when writing their answer. These students should receive full credit.

Question	7	Benchmark(s)	MA.8.NSO.1.6
Difference: 3.06×10^{-3} cubic feet		Recommended Scoring (4 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
		Additional Notes	Consider deducting 1 point if the student does not provide an answer using significant digits.

Question	8	Benchmark(s)	MA.8.NSO.1.5
☐ -4.1×10^{-48} ☐ -11.4×10^{-48} ☐ -4.1×10^{-14} ☐ -1.14×10^{-13} ☐ $-.000000000000041$ ☐ $-.000000000000114$ ☐ $-.0000000000000000000000000000000000000000000000041$ ☐ $-.00000000000000000000000000000000000000000000000114$		**Recommended Scoring** **(8 Total Points)**	Consider giving students 4 points for each correct answer and deduct 2 points for each incorrect answer. A student who selects all of the answers should receive no points, and a student should not receive negative points
Question	**9**	**Benchmark(s)**	MA.8.NSO.1.5
☐ -4.6637×10^{2} ☐ -985.85×10^{17} ☐ -9.8117×10^{19} ☐ -466.37 ☐ $-98{,}585{,}000{,}000{,}000{,}000{,}000$ ☐ $-98{,}117{,}000{,}000{,}000{,}000{,}000$		**Recommended Scoring** **(8 Total Points)**	Consider giving students 4 points for each correct answer and deduct 2 points for each incorrect answer. A student who selects all of the answers should receive no points, and a student should not receive negative points

<table>
<tr><td>Question</td><td>10</td><td>Benchmark(s)</td><td>MA.8.NSO.1.5</td></tr>
<tr><td colspan="2">

Scientific Notation	Standard Form
-1.25×10^{2}	-125

</td><td>Recommended Scoring

(8 Total Points)</td><td>Consider 4 points for the correct scientific notation and 4 points for the correct standard form. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.</td></tr>
<tr><td>Question</td><td>11</td><td>Benchmark(s)</td><td>MA.8.NSO.1.5</td></tr>
<tr><td colspan="2">

Scientific Notation	Standard Form
5.8×10^{-5}	0.000058

</td><td>Recommended Scoring

(8 Total Points)</td><td>Consider 4 points for the correct scientific notation and 4 points for the correct standard form. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.</td></tr>
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Question	12	Benchmark(s)	MA.8.NSO.1.5
<table><tr><th>Scientific Notation</th><th>Standard Form</th></tr><tr><td>5.3×10^{-3}</td><td>0.0053</td></tr></table>		**Recommended Scoring** **(8 Total Points)**	Consider 4 points for the correct scientific notation and 4 points for the correct standard form. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**13**	**Benchmark(s)**	MA.8.NSO.1.5
<table><tr><th>Scientific Notation</th><th>Standard Form</th></tr><tr><td>1.74×10^{13}</td><td>17,400,000,000,000</td></tr></table>		**Recommended Scoring** **(8 Total Points)**	Consider 4 points for the correct scientific notation and 4 points for the correct standard form. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	14	Benchmark(s)	MA.8.NSO.1.5
<table><tr><th>Scientific Notation</th><th>Standard Form</th></tr><tr><td>4.5×10^{-3}</td><td>0.0045</td></tr></table>		**Recommended Scoring** **(8 Total Points)**	Consider 4 points for the correct scientific notation and 4 points for the correct standard form. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.